Progress of Mobile Video Streaming from all Generations Network to in Millimeter Wave 5G Networks

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***Abstract* – In this paper I will through light on evolution of mobile video streaming Networks along with their significance and advantage. In few past decades, the mobile wireless evolution progressed from Zero Generation (0G) to First Generation (1G), Second Generation (2G), Third Generation (3G), and now Fourth Generation (4G) systems are being deployed with the aim Quality of Service (QoS), efficiency and performance. Mobile wireless technology has reached to 4G until the future 5G of Technology. Evolution of mobile, the recent growth in video streaming and the resulting congestion of social network and video-sharing website have increased, as there has been an explosive increase in wireless data traffic. This paper presents the current progress of video streaming, which is fundamental to advanced applications in the fifth generation (5G) networks. Millimeter wave (mmWave) communication represents a leading 5G technology, which provides rich bandwidth and, therefore, great potential for high-quality mobile video streaming.**

***Index Terms* – *Progress, Generations of mobile wireless, Millimeter waves, Fifth generation of mobile communications*.**

I. INTRODUCTION

Video streaming stands for the concept of playing realtime video by retrieving source segments from web video server simultaneously, and it is the most popular online video playing technique [6]. Mobile data traffic increases dramatically with the exponential growth in the number of mobile devices and the emerging of high-rate multimedia applications (such as video streaming for mobile gaming and social networks) [8], while mobile networks of the second and third generations, such as GSM and UMTS, will be still much in use for quite some time to come in many regions of the world, and the deployments of the fourth generation, commonly referred as LTE, are still significantly growing, there is no doubt that work towards the fifth generation, 5G, has gained a lot of momentum. As the NGMN Alliance puts it in its recent white paper, 5G addresses the demands of 2020 and beyond, and “is expected to enable a fully mobile and connected society and to empower socio-economic transformations in countless ways” [9].

Since analog (0G-1G) to digital (2G-5G), the video streaming has taken birth due to digital telecommunications. 0G, also known as Mobile radio telephone, are the systems that preceded modern cellular mobile telephony technology [10]. 1G (or 1-G) refers to the first generation of wireless telephone technology (mobile telecommunications). These are the analog telecommunications standards that were introduced in the 1980s and continued until being replaced by 2G digital telecommunications. The main difference between the two mobile telephone systems (1G and 2G), is that the radio signals used by 1G networks are analog, while 2G networks are digital [10]. 2G (or 2-G) is short for second-generation wireless telephone technology. Second generation 2G cellular telecom networks were commercially launched on the GSM standard in Finland by Radiolinja (now part of Elisa Oyj) in 1991.Three primary benefits of 2G networks over their predecessors were that phone conversations were digitally encrypted; 2G systems were significantly more efficient on the spectrum allowing for far greater mobile phone penetration levels; and 2G introduced data services for mobile, starting with SMS text messages. 2G technologies enabled the various mobile phone networks to provide the services such as text messages, picture messages and MMS (multimedia messages). All text messages sent over 2G are digitally encrypted, allowing for the transfer of data in such a way that only the intended receiver can receive and read it [10]. 3G, short form of third generation, is the third generation of mobile telecommunications technology. This is based on a set of standards used for mobile devices and mobile telecommunications use services and networks that comply with the International Mobile Telecommunications-2000 (IMT-2000) specifications by the International Telecommunication Union. 3G finds application in wireless voice telephony, mobile Internet access, fixed wireless Internet access, video calls and mobile TV. 3.5G is a grouping of disparate mobile telephony and data technologies designed to provide better performance than 3G systems, as an interim step towards deployment of full 4G capability. The technology includes:

- High-Speed Downlink Packet Access
- 3GPP Long Term Evolution, precursor of LTE Advanced
- Evolved HSPA [10].

4G provides, in addition to the usual voice and other services of 3G, mobile broadband Internet access, for example to laptops with wireless modems, to smartphones, and to other mobile devices. Potential and current applications include amended mobile web access, IP telephony, gaming services, high-definition mobile TV, video conferencing, 3D television, and cloud computing. 4.5G is a grouping of disparate mobile telephony and data technologies designed to provide better performance than 4G systems, as an interim step towards deployment of full 5G capability. The technology includes:

- LTE Advanced
- MIMO [10].

5G denotes the next major phase of mobile telecommunications standards beyond the current 4G/IMT-Advanced standards. NGMN Alliance or Next Generation Mobile Networks Alliance define 5G network requirements as:

- Data rates of several tens of Mb/s should be supported for tens of thousands of users.
- 1 Gbit/s to be offered, simultaneously to tens of workers on the same office floor.
- Several hundreds of thousands of simultaneous connections to be supported for massive sensor deployments.

- Spectral efficiency should be significantly enhanced compared to 4G.
- Coverage should be improved.
- Signaling efficiency enhanced.
- Latency should be significantly reduced compared to LTE

Simply providing faster speeds, they predict that 5G networks will also need to meet the needs of new use-cases such as the Internet of Things as well as broadcast-like services and lifeline communications in times of disaster [10].

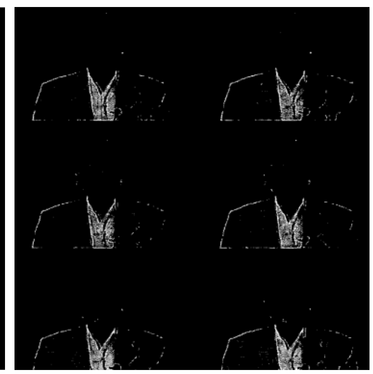
Millimeterwave (mmWave) communication represents a leading 5G technology, which provides rich bandwidth and, therefore, great potentials for high-quality mobile video streaming. However, mobile video streaming in mmWave 5G networks faces fundamental challenges. Future fifth generation (5G) networks need to be designed to accommodate the overwhelming mobile traffic demands. Millimeter wave (mmWave) spectrum with huge available bandwidth is a promising technology for 5G networks to satisfy the high capacity demands and overcome the bandwidth shortage at saturated microwave spectrum. mmWave communication has severe propagation loss because of the high frequency band. Directional antenna is adopted for both mobile users and mmWave base stations to combat the severe propagation loss and achieve high data rate. Connectivity maintaining by directing antenna beams towards each other is challenging, especially for high mobility scenarios (such as highway and rail environments). mmWave base stations are expected to be deployed densely in small cell (e.g., picocells with range under 100 meters) to provide high data rates and aggregate capacity. Users with high mobility would suffer frequent handoffs due to smaller cell size [8].

In line with this purpose, this paper raises a number of questions that need to be addressed in the design of 5G networks about video streaming. While we do not believe that it is already necessary to give final answers to these questions, we feel it is important to make the 5G architects aware of them and start discussion on them [9]. In the remainder of this paper, we first outline our current view of the most important progressing of video streaming from 2G to 4.5G to see how far the evolution mobile technology has progressed (section II). We examine all about mobile video streaming in Millimeter Wave 5G Networks in more detail (section III). We conclude our paper; conclusions are given in Section IV.

II. VIDEO STREAMING FROM 2G TO 4.5G

A progressive video codec can be designed to be finely scalable, that was, it can have produced one fully embedded bitstream which can be decoded at any intermediate rate up to the full target rate, producing progressively better video quality with increased bit-rate. Video scalability is an attractive feature that not only provides bit-rate adaptability to different bandwidth conditions over a heterogeneous network but also provides a framework for introducing graceful degradation in video quality in the presence of bit errors. The priority of the bits in a progressive video bitstream is, by design, well defined. Thus, efficient channel coding schemes can be designed. They proposed an unequal error protection scheme in the application layer for the video bitstream that works in concatenation with the channel codes in the physical layer of state-of-the-art cellular systems and demonstrate that progressive video streaming can be achieved over standard compliant Second Generation (2G) (and beyond) cellular systems without having to modify the physical layer of the systems. The channel protection scheme described would also work for a layered coder or for a coder where one could prioritize the data from the most to the least important. A progressive video coder allows us not only to provide bit-rate adaptability to different bandwidth conditions over a heterogeneous network, without transcoding, but also provides a framework for introducing graceful degradation in video quality in the presence of bit errors. The description of an unequal error protection scheme in the application layer that works in conatenation with the channel codes in the physical layer of standard cellular systems and demonstrate that progressive video streaming, with graceful degradation in video quality in the presence of bit errors, may be realized in standard 2G and 3G wireless systems. Furthermore, if feedback or ARQ is available in the system, due to the fact that a progressive video bitstream is fully scalable, tradeoffs between video and channel code rates can readily be implemented dynamically depending on the channel conditions and bandwidth efficient video transmission scheme can be derived [1].

No Error Detection (left) and error Detection (right) with bit error occurring at 25% of the bitstream [1].



No Error Detection (left) and error Detection (right) with bit error occurring at 75% of the bitstream [1].

The increased capacity provided by the 3G radio access technologies (RAT), combined with the computing power of the smartphones have been resulted in a significant growth in mobile traffic. This growth was mainly fueled by the popularity of large video content on mobiles, which has pushed mobile operators towards providing TV and Video on Demand on their networks. This joint increase in access capacity and user demand affected traffic patterns. As a result, a major concern for operators was to ensure that their mobile network provides a good level of performance when faced with these traffic characteristics. Focusing on the user experience when viewing multimedia content, the evaluation of two content delivery techniques: live TV and progressive download. The view of how their behavior differs and how the Radio Access Type influences their performances cab be seen [2]:

• Live TV is a popular service offered by the ISP whose bitrates are standardized and well adapted to the radio access. As the loss rate is low, the playback buffer should allow a good video quality apart of some loss bursts.

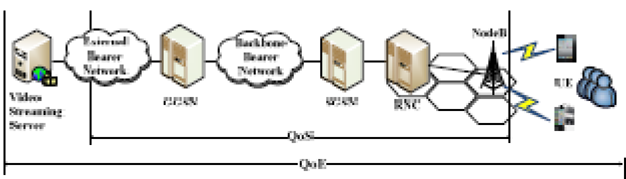
• Progressive download allows users to begin watching the content after a short buffering duration. In order to assess the user experience, we propose a method for estimating the number of playback interruptions from passive network measurements. We observe that 3G users generally enjoy an interruption-free playback as the actual network throughput is usually higher than the content encoding rate. However, most 2G users can conveniently retrieve audio streams only. We also show that PDL users react to bad playback quality by prematurely aborting their sessions.

The lessons of these measurements are that:

• HSDPA users could be provided higher encoding rates for videos;

• 2G users should mainly be proposed audio contents and shorter streams, if possible [2].

With the development of 3G networks and the emergence of intelligent terminals, more and more people enjoy the video streaming service through 3G networks and portable terminals. Since the video streaming service demands a higher resource than traditional services, such as voice and web browsing, how to guarantee the user experience is a challenge issue to the network operator. Quality of experience (QoE), defined as "*the overall acceptability of an application or service, as perceived by the end-user*" in [1], not only includes the quality of service (QoS), but also considers the capability of the user equipment (UE) as well as user's expectation and context. Hence, QoE is a comprehensive indicator to measure the performance of the end-to-end systems, while QoS only considers the network parameters, such as packet loss rate, delay, and so on [3]. As shown in Figure.



[3]

Consumption of multimedia on mobile devices is increasing; YouTube alone accounts for 22% of the mobile data bandwidth. Moreover, the video streaming services cannot only be accessed by smartphones and tablets but also by constrained devices like a feature phones and in-vehicle entertainment systems. In a 3G network, mobility, fading, interference, cell loading, handovers and other factors can affect the throughput available to each user. Generally, video streaming applications overcome congestion by prebuffering content. However, the video will pause if the buffer runs out and will not resume until a sufficient amount of video is pre-buffered [4]. Next-generation wireless networks are expected to support a wide variety of high-speed data applications, including increasingly popular streaming applications. The integration of these heterogeneous applications on a common transmission infrastructure raises major challenges for the deployment of opportunistic schemes. By examining a power utility function that captures the user-level performance in a mixed service [5].

Fourth Generation wireless systems was expected to extend wireless service to high data rates in high-mobility Environments [7] but The increasing adoption of mobile 4G LTE networks, video streaming as the major contributor of 4G LTE data traffic, has become extremely hot. However, the battery life has become the bottleneck when mobile users are using online video services [6]. the measurement and modeling of smartphone power consumptions of video streaming in 4G LTE networks. The presentation of a compound platform to collect the power, throughput and RRC state data on mobile phones [6]. With exponential increase in the volumes of video traffic in cellular networks, there was an increasing need for optimizing the quality of video delivery. 4G networks (Long Term Evolution – Advanced or LTE-A) were being introduced in many countries worldwide, which allow a downlink speed of upto 1 Gbps and uplink of 100 Mbps over a single base station [12] the performance of LTE-A physical layer in terms of transmitted video quality when the channel conditions and LTE settings are varied. The test of the performance achieved as the channel quality is changed and HARQ features are enabled in physical layer. Blocking and blurring metrics were used to model image quality. Mobile video is quickly becoming a mass consumer phenomenon; much as digital photos were earlier in the smartphone adoption cycle. There is a tectonic shift to ondemand and on-the-go video services. Video streaming services such as Netflix and Youtube are generating large volumes of traffic and revenue (3 billion $ for Netflix in 2011) [12]. the cellular networks were migrating towards a new technology 4G LTE which is 10x faster than 3G. Boosted bandwidth allows for quicker uploads and better streaming quality. LTE-A targets to achieve peak uplink and downlink data rates of 500 Mbps and 1 Gbps, respectively, for low-speed UEs and around 100 Mbps for those with higher mobilities. It accommodates the next generation of telecommunication services such as realtime high-definition video streaming, mobile HDTV and high quality video conferencing. In LTE-A systems, the bandwidths in both uplink and downlink can go upto 100 MHz, which is achieved by Carrier Aggregation (CA) or aggregation of individual Component Carriers There have been volumes of research in different aspects of multimedia systems, to enhance the performance of mobile video. Hardware architectures have been proposed for mobile video compression and encryption. Efficient video transmission in cellular low bandwidth scenarios has also been discussed. With 4G LTE becoming popular, we attempt to understand the features of LTE which help in high quality video transmission. The major features that distinguish LTE from 3G technologies at the air-interface are Orthogonal Frequency Multiple Access (OFDMA), advanced MIMO technology, and Hybrid Automatic Repeat Request (HARQ). In addition, LTE uses flat-IP architecture for the core network. LTE uses OFDMA in the downlink (DL) for efficient multiple-access and for countering multipath frequency selective fading. OFDMA divides the available channel into number of sub-carriers and is naturally suitable for scalable bandwidth allocation by varying the FFT (Fast Fourier Transform) size. LTE uses SC-FDMA (Single Carrier FDMA) in the uplink (UL) to obtain a low peak-to-average-power ratio (PAPR) [12].

Ⅲ. MOBILE VIDEO STREAMING IN MILLIMETTER WAVE 5G

Millimeter wave (mmWave) spectrum with huge available bandwidth is a promising technology for 5G networks to satisfy the high capacity demands and overcome the bandwidth shortage at saturated microwave spectrum. mmWave communication has severe propagation loss because of the high frequency band. Directional antenna is adopted for both mobile users and mmWave base stations to combat the severe propagation loss and achieve high data rate. Connectivity maintaining by directing antenna beams towards each other is challenging, especially for high mobility scenarios (such as highway and rail environments). mmWave base stations are expected to be deployed densely in small cell (e.g., picocells with range under 100 meters) to provide high data rates and aggregate capacity. Users with high mobility would suffer frequent handoffs due to smaller cell size. Mobile video streaming is one of the main applications for 5G networks. Enabling mobile video streaming in mmWave 5G networks has several challenges in high-mobility environments. First, due to directional antenna, the connectivity from mmWave base station to mobile user can be available to deliver video content only if both antennas direct towards each other. Each mobile user has very short connection time to the base station considering high user mobility, antenna directivity, smaller cell size, and the fact that a larger number of users share the mmWave channel in time division. Second, with smaller cell size, users with high-mobility have frequent handoffs and need to frequently re-build the route to remote video server through different base stations connected to the core network, which involves heavy communication overheads and long latency. The delay for re-building the route to deliver video content has significant impact on the video streaming quality, such as video frozenness. In this paper, to provide satisfactory quality for mobile video streaming requiring high data rate (e.g., uncompressed video streaming requires mandatory data rate of 1.78/3.56 Gbps in 5G networks), a proactive caching system is enabled by preloading each user's video content from the remote video server and storing the video content in the cache memory of each base station. A mobile user entering a new cell can immediately have the available video content, which mitigates the delay on frequently re-building the route to remote video server. Larger size of pre-loaded video content for a specific user in the cache memory can provide better video streaming quality while including more video delivery cost and occupying more cache memory space which can be used by other users. Smaller size of pre-loaded video content might not be sufficient for the user to maintain video streaming quality until the next video content is received. Therefore, how to optimize the size of preloaded video content in each user's subsequent base station is an important issue to improve the quality of mobile video streaming. Quality provisioning for video streaming has been intensively studied in wireless networks, subject to wireless bandwidth limitation and wireless channel fluctuation. One category of research focuses on efficient bandwidth utilization to provide the required quality of service (QoS) of video streaming [8]. In [13], an economic model is used to allocate radio resource between cellular network and wireless local area network (WLAN), aiming to optimize the total welfare of the heterogeneous networks, The wireless channel fluctuation effect on video streaming quality by bufferinga certain amount of video data at end user [8]. In [14], the impacts of wireless channel dynamics on video streaming quality is analytically investigated by modeling the receiver buffer with G/G/1/M queue and G/G/1/N queue. mobile video applications (e.g., Youtube) periodically download video data aggressively into end user's buffer without considering buffer status and network resource availability, in order to reduce video frozenness. An intelligent cost aware buffer management strategy for mobile video streaming applications is proposed to minimize the cost resulting from un-consumed video data while respecting certain quality of experience (QoE) requirements. Literature [14] jointly considers the bandwidth allocation and buffer management at the mobile user to dynamically charge/discharge the buffer to optimize video streaming quality. The challenges in delivering high-demanding multimedia applications especially UHD videos in 5G mobile networks, the proposition of a QoE and context-aware system. The latest generation of video codecs, H.265 and its extensions are explored, and an SDN-controlled framework is proposed. The proposition of VQAM builds upon our previous work in scalable video streaming in SDN environments with more built-in intelligence in the areas of better link/path utilization for multiple layered streams prior to congestion, and a set of decisive steps in conjunction with the SDN-VQO to continue to deliver the required QoE in the event of a network congestion. In our proposed framework, we aim to harness the finegrained scalability features of the latest video codecs, and the flexibility SDN brings to network architectures to help tackle some of the challenges envisaged or imminent in the future of mobile multimedia networks with an emphasis on QoE and meeting 5G's KPIs. Another feature that could be built into this system is a distributed framework, with periodic coordination via updates between controllers to reactively facilitate the management of localized congestions via a wider view of the entire network. We have currently implemented and validated many of the individual components of the proposed VQAM related to video adaptation and are currently working towards implementation of the SDN-VQO [8].

IV. Conclusion

This paper focuses on the progress of mobile video streaming through all generations of network and in the future Millimeter Wave 5G. In this paper, we can see the evolution of video streaming from the bottom, in the present and in the future. The video streaming really started with 2G then 3G and so on. We tried to touch all the network generations, giving some advantages, weaknesses and improvements. Some information about the millimeter wave 5G, hence the video streaming of mobile in the generations of network have progressed considerably and will be even more better in the 5G.

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